SH E.656

"Caimano"

SH E.656 Three-Phase Electric Locomotive

SILVER HIVE

v1 Expert

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1. Introduction

This manual describes the operation of the **SH E.656 'Caimano'** for Train Simulator Classic (TSC), developed by **Silver Hive (SH)**. Its aim is to provide a thorough and accurate guide for correct operation of the locomotive, from service preparation through to shutdown, including full management of the SCMT safety system.

This manual is intended for TSC players and assumes a basic familiarity with the simulator. For detailed technical documentation of the SCMT system, refer to the official DLC manual by Giorgio Musciagna (Worcestergeorge), available at worcestergeorge.altervista.org.

■ Expert Edition
This is the full Expert version of the SH E.656. It includes all advanced procedures: PAC manual rheostat override, SHUNT field weakening, IR protection logic, and the complete SCMT system with SSB-SCMT start-up and data entry.

2. Locomotive Overview

2.1 Real Prototype

The E.656, affectionately nicknamed 'Caimano' (Caiman) for its distinctive low, elongated nose, is one of the most iconic electric locomotives in Italian railway history. Designed between 1973 and 1975 and built in 461 units through to 1989, it served for decades as the backbone of heavy traction on the Italian State Railways.

Figure 1 — SH E.656 'Caimano' — Side/front view

2.2 Technical Specifications

Parameter	Value
Type	Electric locomotive
Power supply	3000 V DC
Continuous power	3480 kW (4 motors × 870 kW)
Maximum speed	160 km/h
Service mass	~83 t
Wheel arrangement	Bo'Bo'Bo'
Gear ratio (E.656)	28/61
Gear ratio (E.655)	23/66
Cab configuration	Bi-cab
Safety system	SCMT + RSC (Worcestergeorge DLC)

Parameter	Value
Mechanical manufacturer	Casaralta · OMR · SOFER · TIBB
Electrical manufacturer	Ansaldo · Ercole Marelli · Asgen · Italtrafo
Production period	1975–1989
Units built	461

2.3 Enabling Providers in TSC

Before running a scenario with the E.656 ‘Caimano’, both the Silver Hive and Worcestergeorge providers must be enabled in the TSC editor. The procedure applies equally to the locomotive and to any support vehicles in the scenario.

Step 1 — Select the SilverHive Provider

In the TSC editor, click the **blue square** (circled in red) in the upper-left corner. From the drop-down menu on the right select **SilverHive**: the entry **SH_E656_SCMT** will appear in the available rolling stock list.

Figure P1 — Selecting the SilverHive provider in the TSC editor

Step 2 — Enable the SCMT Plugin (Worcestergeorge)

Click the blue square again and select the **Worcester-George** provider from the drop-down. **Tick all the checkboxes** as shown, in particular the **SCMT** entry. Repeat for the locomotive itself.

Figure P2 — Worcester-George provider: SCMT checkbox enabled

■ Caution — SCMT Plugin Required
If the Worcester-George provider is not activated or its checkboxes are not ticked, the SCMT will not function correctly and the traction lever will remain locked. Ensure the Worcestergeorge SCMT DLC is installed before running any scenario.

3. Cab Overview

The locomotive has two identical driving cabs, referred to as Cab A (front) and Cab B (rear) relative to the current direction of travel. The main controls are described in the sections below.

3.1 Cab A (Front)

Figure 2 — General view of Cab A (driver's desk)

3.2 Cab B (Rear)

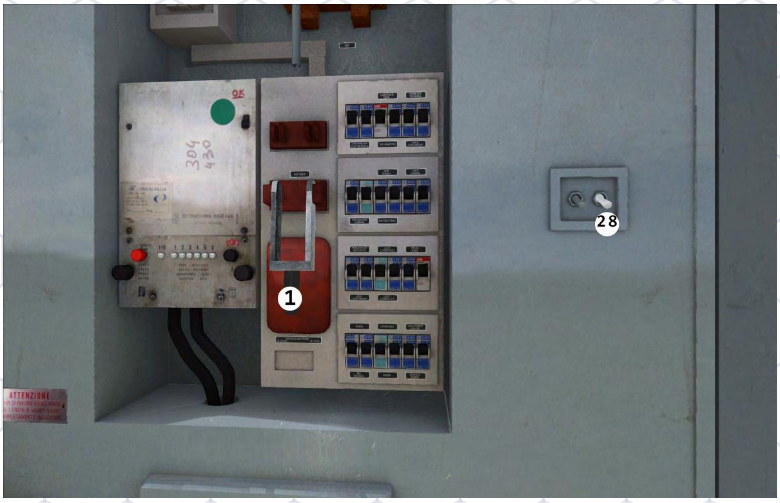

Figure 3 — Left: switch and fuse panel | Right: CPA pneumatic system

3.3 Main Components

No.	Component	Description
1	Battery (Bipolar)	Main battery switch — closes the primary circuit
2	Pantographs T1/T2	Pantograph raise/lower controls

No.	Component	Description
3	Command circuit switch	Enables the control circuit (Cmd Circuit)
4	CPA	First-raise compressor — builds pressure for pantograph
5	CPA cock	Rotary cock for CPA circuit isolation
6	Bench key	Cab enabling key (blind)
7	IR (Fast Circuit Breaker)	Closes/opens the main IR — black/red buttons
8	STROTZ (Static Groups)	Auxiliary static power groups (lighting, heating)
9	MC1 / MC2	Motor-compressors 1 and 2 — main air supply
10	Motorised fans	Cooling fans for resistances and traction motors
11	RAP	Departure acknowledgement — vigilance device button
12	RAE potentiometer	Adjusts the rheostat response speed
13	Reverser	Direction selector — Forward / Neutral / Reverse
14	Combination selector	Traction mode selector (S, SP, P, PP, M positions)
15	PAC	Manual rheostat override — active in M (Shunting) only
16	SHUNT	Field weakening — increases max speed beyond current notch
17	Graduated brake	Locomotive-only brake, fully modulable
18	Continuous brake	Train brake — Westinghouse air brake on entire consist
19	Brake isolating cock	Isolates the continuous brake circuit (yellow lever)
20	Reservoir valve	Main air reservoir isolating valve
21	Spring brake (HandBrake)	Parking/spring brake handwheel
22	Sander	Manual sander — improves wheel adhesion
23	Horn	Acoustic warning device

4. Service Preparation

The preparation sequence is divided into four phases. Complete each phase in order and wait for confirmation before proceeding.

PHASE 1 — ON-BOARD CIRCUIT

4.1 Activating the On-Board Circuit

1. **Engage the Battery — key [B].**
   Lift the lever upward to close the primary circuit. Wait ~10 seconds.
2. **Enable the Command Circuit — key [SHIFT+O].**
   Ensure the switch is set upward (enabled).
3. **Start the CPA (First-Raise Compressor) — key [C].**
   The auxiliary compressor builds pressure for the pantograph. Target: 6 bar.

4	**Rotate the CPA Cock — key [SHIFT+C].** Rotate the cock lever to feed air to the pantograph circuit.
5	**Raise Pantograph T1 — key [SHIFT+G].** Wait for the pantograph to lock against the overhead wire.
6	**Raise Pantograph T2 — key [SHIFT+T].** Both pantographs raised for redundant current collection.
7	**Close the IR (Fast Circuit Breaker) — key [Y].** Press the black button. The IR closes and the traction circuit is energised.
8	**Enable STROTZ (Static Groups) — key [J].** Powers the auxiliary systems: lighting, heating, SCMT.

PHASE 2 — COMPRESSORS AND FANS

4.2 Starting Auxiliary Groups and Compressors

9	**Start Motor-Compressor MC1 — key [K].** MC1 begins charging the main air reservoirs.
10	**Start Motor-Compressor MC2 — key [SHIFT+K].** Both compressors running. Wait for main reservoir pressure ≥ 8 bar.
11	**Enable Automatic Motorised Fans — key [L].** Cooling fans for traction motor resistances come online automatically.

PHASE 3 — BRAKING SYSTEM

4.3 Enabling the Braking System

Figure 4 — Brake column: isolating cock (19), continuous brake (18), graduated brake (17) and reservoir valve (20)

12	**Open the Brake Isolating Cock — key [SHIFT+M].** Yellow lever: rotate to the open position to enable the brake circuit.
13	**Lift the guard and release the Continuous Brake — keys [and].** Lift the guard on the brake handle first, then move it to the release position.
14	**Enable the SCMT Switch — key [N].** Wait for the self-test to complete. Refer to the Worcestergeorge SCMT manual for details.
15	**Release the Spring Brake (HandBrake) — key [Backspace].** Confirm the parking brake is fully released before traction.

PHASE 4 — TRACTION

4.4 Starting Traction

16 **Insert the Bench Key — key [O].**
Enables the cab (lowers the blind). The desk is now active.

17 **Set the direction with the Reverser (13) — Forward or Reverse.**
Select using the direction keys. Ensure the reverser is not in Neutral.

18 **Move the Combination Selector (14) to position S (Series).**
Traction builds progressively through the rheostat. Advance gradually to avoid wheel-slip.

■ **Caution — RAP (Vigilance Device)**
At 2 km/h the RAP vigilance device activates. Press the RAP button (11) within 2.5 seconds to confirm driver presence. Failure to press triggers automatic emergency braking.

4.5 PAC — Manual Rheostat Override

The **PAC** lever allows manual control of the starting rheostat. It is operative **exclusively in position M** (Shunting) of the combination selector. Each pulse advances the rheostat target by **one notch** (scale 0–31). The lever is spring-returned: it resets to zero after approximately 0.4 seconds.

Condition	PAC Operative	Notes
Selector in position M	✔ Yes	Manual notch-by-notch control
Selector in S / SP / P / PP	✘ No	Rheostat automatic via RAE
IR open or line absent	✘ No	No effect on traction
Selector in 0 (Neutral)	✘ No	

■ **Note — Thermal Protection Reset**
Each PAC input resets the thermal protection timers (‘rheostat dwell’ and ‘motor overload’), allowing extended traction sessions in Shunting mode.

4.6 SHUNT — Field Weakening

SHUNT weakens the traction motor magnetic field, reducing back-EMF and allowing the locomotive to exceed the maximum speed of the current combination. All of the following prerequisites must be met **simultaneously**; otherwise the input is ignored.

Prerequisite	Required State
Rheostat	Fully excluded (notch 31/31)
Speed	≥ 30 km/h
IR	Closed, no active trip
Line	Active (voltage present)
Selector	S, SP, P or PP (not M, not Neutral)

> **■ Caution — Auto-Reset of SHUNT**
> SHUNT resets automatically if: speed drops below 25 km/h, the rheostat is no longer fully excluded, the line deactivates, or the IR trips.

4.7 IR Protections — Trips and Reset

The Fast Circuit Breaker (IR) is protected by five automatic trip logics. When a trip occurs, a **pop-up appears in the top-right corner** of the screen with the title **'SH E.656 — IR TRIP'** and the specific cause. Traction is cut immediately.

Trip Causes

Cause	Threshold / Condition	Delay
Branch overcurrent	Branch current ≥ 650 A	3 seconds
Rheostat dwell	Rheostat engaged (< notch 31) with current active	80 seconds
Motor overload	Current ≥ 85% I_{nom} with rheostat excluded	80 seconds
Downward shift under load	Steps down combination with current > 120 A	Immediate
Violent combination jump	Skips > 1 combination at v > 30 km/h	Immediate
IVR thermal trip	Fan temperature ≥ 255°C	Immediate

IR Reset Procedure

1. **Return the combination selector to position 0 (Neutral).**
   Reset is blocked if the selector is not in position 0.
2. **Confirm that the cause of the trip has cleared.**
   For IVR thermal trips, wait for fan temperature to drop below 201°C.
3. **Close the IR — key [Y].**
   The pop-up 'IR Reset Successfully' will appear if the reset is accepted.
4. **Resume traction from position S.**
   Re-engage the reverser, then advance the selector to S and increase gradually.

> **■ Caution — Reset Denied**
> If the IVR (fans) has reached a temperature above 201°C, the pop-up 'IR Locked: IVR above 201°C' prevents reset. Allow natural cooling before retrying.

5. Driving

5.1 Start and Departure

1. **Confirm the SCMT system is in the correct operating mode (e.g. PredCMT).**
   The MMI display must show the active mode. Refer to Section 6 for details.
2. **Set the direction of travel.**
   Use the reverser in the cab.
3. **Release the parking brake — key [Backspace].**

4	**Release the automatic brake gradually.** Move the brake cock to the running position.
5	**Increase traction gradually — key [A].** Advance progressively through the notches to avoid wheel-slip.

5.2 Traction

The traction controller manages the power applied to the motors. Always increase gradually to prevent wheel-slip and to avoid triggering the IR overload protection. Use **M (Shunting)** for yard movements, **S / SP / P / PP** for line running in progressive order.

5.3 Braking

Reduce traction to zero before applying brakes. The **continuous brake (18)** acts on the entire consist via the Westinghouse air brake. The **graduated brake (17)** acts exclusively on the locomotive axles and is fully modulable. Apply the **spring brake (21)** only with the locomotive stationary.

5.4 Lights

Key	Function	Notes
H	Front/rear headlights	
SHIFT+H	Second headlight bank	
CTRL+H	Third light	
E	Cab light	
SHIFT+E	Driver’s light	
CTRL+E	Guard’s light	
SHIFT+F	Instrument lights	

5.5 Auxiliary Systems

System	Key	Notes
Manual sander	U	Improves wheel adhesion on slippery rail
Horn (low)	—	
Horn (high)	Space	
Windscreen wiper	V	

6. SCMT — Train Control System

6.1 Introduction

The SCMT (Sistema Controllo Marcia Treno — Train Movement Control System) is the safety system that monitors train operation, enforcing compliance with maximum permitted speeds, signal aspects, and correct RSC management. The E.656 ‘Caimano’ integrates the **FS SCMT DLC v3.1**, developed by Giorgio Musciagna (Worcestergeorge).

The description below is a gameplay-oriented summary. For the full technical treatment, refer to the official DLC manual.

> **■ Note — Worcestergeorge SCMT DLC**
> The SCMT system in this locomotive is provided by the free Worcestergeorge SCMT DLC. It must be installed and activated in the TSC editor as described in Section 2.3. Download: worcestergeorge.altervista.org/dlc

6.2 SSB-SCMT Start-up

The SSB-SCMT (on-board SCMT unit) starts automatically when the SCMT switch is enabled (step 13 of Section 4.3). During start-up it performs a self-test; the MMI display shows the test progress. Once complete, the system enters the initial **Attesa** (Standby) state.

State	MMI Display	Description
Attesa	Empty screen	System powered; BM key not yet inserted
PredCMT	Empty screen	Awaiting first SCMT track circuit (PI)
CMT	Current time	SCMT function active — full supervision

6.3 Data Entry

Before departure, enter the consist data in the **Data Entry** state. Press the **Dati** button on the MMI to access the list.

Data field	Default	Values	Description
Locomotive use	In testa	In testa / Comp AP / Spinta MS	Loco position in consist
PMF	50	45 … 160	Braked mass percentage
Brake type	G	G / P	G = freight, P = passenger
Train type	V	V / M	V = passenger, M = freight
Max train speed	30	30 … 300 km/h	Consist maximum speed
Range	A	A / B / C / P	Operating range
Mass	20	20 … 2000 t	Consist mass

6.4 Operating Modes

The SSB-SCMT operates in several modes depending on line equipment and driver input. The most common are PredCMT (pre-supervision, awaiting first PI), CMT (full SCMT supervision with PI data), and CMT+RSC (SCMT with RSC active). Refer to the Worcestergeorge manual for the complete state diagram.

6.5 Signal Management

As the train passes each SCMT track circuit (PI), the SSB-SCMT receives telegrams containing signal aspect, line speed, and approach information. The CV (speed supervisor) generates a speed profile that the train must respect. If the speed alert curve is exceeded, an intermittent audible warning sounds and traction is cut; if the control curve is exceeded, automatic emergency braking is applied.

6.6 RSC — Continuous Signal Repetition

On lines equipped with RSC (BAcc track circuits), the SSB-SCMT receives continuous speed codes from the rails. These are displayed on the MMI desk panel. When the RSC is active, the **RSC button** on the MMI glows with a steady blue light.

Code	Icon	Meaning
270**	SV	Clear for ≥ 5400 m
270*	VM	Clear for ≥ 4050 m
270	Green	Clear for ≥ 2700 m
180	White	Caution: speed reduction or signal at reduced distance ahead
120	RV	Clear signal for diverging route
75	Orange	Next signal at Danger at ≥ 900 m
AC	AC	No code (diverge, occupied track)

6.7 SR — Signal Override (Supero Rosso)

The SR function allows legitimate passage of a signal at Danger (e.g. on instruction from the Traffic Controller). It must be activated **only with the train at a standstill**.

1. **Bring the train to a complete stop in front of the signal at Danger.**
2. **Press the SR button on the MMI — key [M].**
   The SR button illuminates red for up to 60 seconds (max 2 min at reduced speed). The train may then proceed past the signal at no more than 30 km/h.
3. **Pass the signal and resume normal operation.**
   SR cancels automatically once the signal has been cleared.

> **■ Caution — SR Misuse**
> Activating SR at speed or without authorisation constitutes a safety violation. Use only when instructed by the Traffic Controller or dispatcher.

6.8 Emergency Braking and RF

The SSB-SCMT applies automatic emergency braking (RF — Frenatura di emergenza) whenever the control speed curve is exceeded, or when the PRE/RIC/vigilance acknowledgement times expire. To resume operation after an RF event:

1. **Reduce speed to zero and wait for the train to stop completely.**
2. **Press the RF acknowledgement button — key [Q].**
   The MMI will confirm the reset. Traction cut is released.
3. **Press RIC to acknowledge any pending RSC code — key [Z].**
   Resume normal traction once the MMI returns to a normal operating mode.

7. Keys and Controls

Summary of all default key bindings extracted from the E_656.xml file. Bindings can be customised through the Train Simulator Classic settings.

System	Key	Function	Notes
Circuit	SHIFT+O	Command Circuit ON/OFF	
	B	Battery (Bipolar) ON/OFF	
CPA / Pantographs	C	CPA — First-Raise Compressor	
	SHIFT+C	CPA Cock	
	O	Bench Key (blind)	
	SHIFT+G	Pantograph T1 (front)	
	SHIFT+T	Pantograph T2 (rear)	
IR / Aux	Y	Close IR	Black button
	SHIFT+Y	Open IR	Red button
	J	Static Groups (STROTZ)	
Compressors	K	Motor-Compressor MC1	
	SHIFT+K	Motor-Compressor MC2	
	L	Automatic Fans	
Brakes	SHIFT+M	Brake Isolating Cock	Yellow lever
	[/]	Graduated Brake — apply / release	
	; / '	Continuous Brake — apply / release	
	Backspace	Spring Brake / Emergency Brake	
Traction	A / D	Combination Selector — up / down	
	W / S	Reverser — Forward / Reverse	
	Q	PAC (rheostat override)	
	Z	SHUNT	
Lights	H	Front/rear headlights	
	SHIFT+H	Second headlight bank	
	CTRL+H	Third light	
	E	Cab light	
	SHIFT+E	Driver’s light	
	CTRL+E	Guard’s light	
	SHIFT+F	Instrument lights	
SCMT	N	SCMT Switch ON/OFF	
	X	SCMT Pre-announcement (PRE)	
	Q	SCMT Emergency Brake Ack. (RF)	
	Z	SCMT Acknowledgement (RIC)	

System	Key	Function	Notes
	M	SCMT Signal Override (SR)	
Auxiliary	U	Manual Sander	
	Space	Horn (high)	
	V	Windscreen Wiper	
	RAP	Vigilance Device Button	

8. Shutdown

1. **Bring the train to a complete stop and apply the parking brake — key [Backspace].**
2. **Release traction by returning the controller to zero.**
3. **Lower the pantographs — keys [SHIFT+T] then [SHIFT+G].**
4. **Open the IR — key [SHIFT+Y].**
5. **Shut down all auxiliary systems (compressors, lights).**
6. **Remove the bench key — key [O].**
   The SSB-SCMT will automatically enter the Standby state.

9. Troubleshooting

Problem	Likely Cause	Solution
Pantograph will not raise	No line voltage in scenario	Check scenario settings and confirm line electrification
SCMT will not start	BM key not inserted or CG pressure too low	Insert the key and wait for CG ≥ 4.5 bar
Sudden emergency braking	Speed alert curve exceeded by CV	Reduce speed and press RF [Q] when the train has stopped
RIC button flashing	Restrictive RSC code received	Press RIC [Z] within 3 seconds
Error on MMI display	Various causes (see SCMT error table)	Execute RF, then press RIC to acknowledge
Loco will not move	Traction cut active (SCMT)	Check MMI state and reset the traction cut

10. Credits and Licence

DLC Author	Silver Hive (SH)
Version	v1 Expert — 2025

Special Thanks

Special thanks to **Giorgio Musciagna (Worcestergeorge)** for developing and making freely available the FS SCMT DLC — a faithful simulation of the Italian railway train control system. Without this extraordinary contribution, it would not be possible to offer an authentic driving experience of the E.656 'Caimano'.

Third-Party Credits

Resource	Author	Reference
FS SCMT DLC v3.1	Giorgio Musciagna (Worcestergeorge)	worcestergeorge.altervista.org/dlc
SCMT Support Forum	Worcestergeorge	rotabili-italiani.org · rail-sim.de

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